

Independent Random variables :-

If (x, y) is two dimensional R.V and then

then $\boxed{P(X=x, Y=y) = P(X=x) \cdot P(Y=y)}$ for discrete R.Vs

and $\boxed{f(x, y) = f(x) \cdot f(y)}$ for continuous R.Vs

conditional distribution, when x & y are independent R.Vs :-

$$P(X=x | Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$

$$= \frac{P(X=x) \cdot P(Y=y)}{P(Y=y)}$$

ie $\boxed{P(X=x | Y=y) = P(X=x)}$ for discrete R.Vs

Similarly for continuous random variables

$$f_{Y/X}(y/x) = \frac{f(x, y)}{f(x)} = \frac{f(x) \cdot f(y)}{f(x)}$$

ie $\boxed{f_{Y/X}(y/x) = f(y)}$ for cont. R.Vs

Q:- The random variable x and y have the following joint density function

$$f(x, y) = \begin{cases} 6(1-x-y), & x > 0, y > 0, x+y < 1 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Find marginal distribution of x & y .
 (ii) are x and y independent.

Soln:- Marginal distribution of x

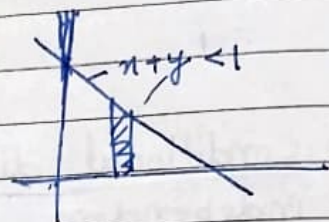
$$f_x(x) = \int_0^{1-x} 6(1-x-y) dy$$

$$= 6 \int_0^{1-x} (1-x-y) dy$$

$$= 6 \left[(1-x)y - \frac{y^2}{2} \right]_0^{1-x}$$

$$= 6 \left[(1-x)(1-x) - \frac{(1-x)^2}{2} \right]$$

$$\left[f_x(x) = 3(1-x)^2 \right]$$

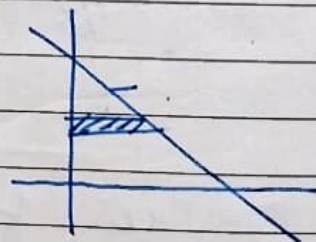


Marginal distribution of y :-

$$f_y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$= \int_0^{1-y} 6(1-x-y) dx$$

$$f_y(y) = 3(1-y)^2$$



check $f_x(x) \cdot f_y(y) = 3(1-x)^2 \cdot 3(1-y)^2$
 $= 9(1-x)^2 (1-y)^2$

$$\neq f_{xy}(x, y)$$

ie x and y are not independent.

2. For r.v X and Y , the joint P.d.f. is given by

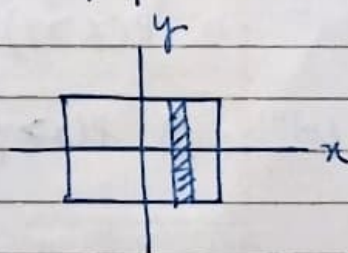
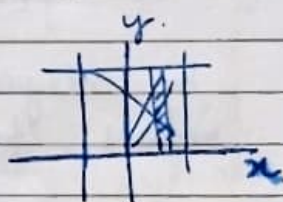
$$f_{xy}(x, y) = \begin{cases} (1+xy)/4, & |x| \leq 1, |y| \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find the marginal density $f_x(x)$, $f_y(y)$ and $f_{y/x}(y/x)$. Are x and y independent

Soln $f_x(x) = \int_{-1}^1 f(x, y) dy$

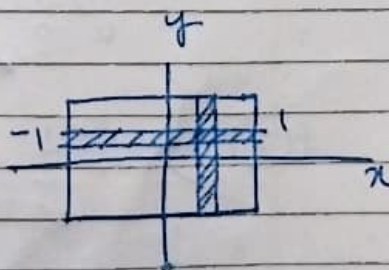
$$= \int_{-1}^1 \left(\frac{1+xy}{4} \right) dy$$

$$f_x(x) = \frac{1}{2}, \quad -1 \leq x \leq 1$$



$$f_y(y) = \int_{-1}^1 f(x, y) dx$$

$$= \frac{1}{2}, \quad -1 \leq y \leq 1$$



$$f_{y/x}\left(\frac{y}{x}\right) = \frac{f_{xy}(x, y)}{f_x(x)}$$

$$= \frac{\frac{1+xy}{2}}{\frac{1}{2}} = 1+xy$$

$$\neq f_x(x) \cdot f_y(y)$$

$$\boxed{f_{y/x}(y/x) \neq f_x(x) \cdot f_y(y)}$$

x and y are not independent.

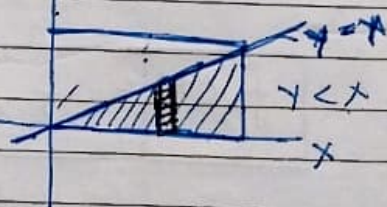
eg:- let x and y have the joint density
 $f(x, y) = \frac{6}{7}(x+y)^2, 0 \leq x \leq 1, 0 \leq y \leq 1$.

by integrating over the appropriate regions, find

- (1) $P(x > y)$ (2) $P(x+y \leq 1)$ (3) $P(x \leq 1/2)$

Soln:- ① $P(x > y) = \iint_R f(x, y) dy dx$

$$= \int_0^1 \int_0^x \frac{6}{7}(x+y)^2 dy dx$$



$$P(x > y) = \frac{1}{2}$$

$$(2) \quad P(X+Y \leq 1) = \iint_R \frac{6}{7} (x+y)^2 dx dy$$

$$= \int_{y=0}^1 \int_{x=0}^{1-y} \frac{6}{7} (x+y)^2 dx dy$$

$$= \frac{6}{7} \int_{y=0}^1 \int_{x=0}^{1-y} (x^2 + y^2 + 2xy) dx dy$$

$$= \frac{6}{7} \int_{y=0}^1 \left(\frac{x^3}{3} + y^2 x + x^2 y \right) \Big|_0^{1-y} dy$$

$$= \frac{6}{7} \int_{y=0}^1 \left[\frac{1}{3} (1-y)^3 + y^2 (1-y) + (1-y)^2 y \right] dy$$

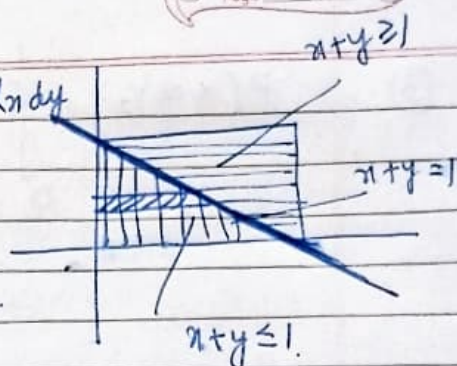
$$= \frac{6}{7} \int_0^1 \left[\frac{1}{3} (1-y)^3 + (y^2 - y^3) + y(1+y^2 - 2y) \right] dy$$

$$= \frac{6}{7} \int_0^1 \left[\frac{1}{3} (1-y)^3 + y - y^2 \right] dy$$

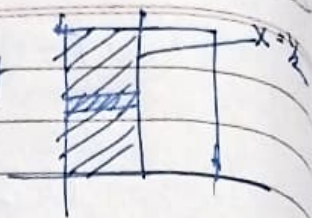
$$= \frac{6}{7} \left[-\frac{1}{3} \frac{(1-y)^4}{4} + \frac{y^2}{2} - \frac{y^3}{3} \right] \Big|_0^1$$

$$= \frac{6}{7} \left[-\left(\frac{1}{12} \times 0 - \frac{1}{12} \right) + \left(\frac{1}{2} - 0 \right) - \left(\frac{1}{3} - 0 \right) \right]$$

$$= \frac{6}{7} \left[\frac{1}{12} + \frac{1}{2} - \frac{1}{3} \right] = \frac{6}{7} \left[\frac{1+6-4}{12} \right] = \frac{6}{7} \times \frac{3}{12} = \frac{3}{14}$$



$$(3) \quad P(X \leq Y_2) = \int_0^1 \int_0^{Y_2} \frac{6}{7} (x+y)^2 dx dy$$



eg:- The joint P.d.f of two dim. R.V is given by $f(x,y) = xy^2 + \frac{x^2}{8}$, $0 \leq x \leq 2$, $0 \leq y \leq 1$

compute

- ① $P(X > 1)$ ② $P(Y < 1/2)$ ③ $P(X > 1 | Y < 1/2)$
 ④ $P(Y < 1/2 | X > 1)$ ⑤ $P(X < Y)$ ⑥ $P(X+Y \leq 1)$

Solution ① $P(X > 1) = \iint_R f(x,y) dx dy$

$$= \int_0^1 \int_1^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy$$

$$= 19/24$$

② $P(Y < 1/2) = \iint_R f(x,y) dx dy$

$$= \int_0^{1/2} \int_0^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy = \frac{1}{4}$$

$$\textcircled{3} \quad P(x > 1, y < y_2) = \int_{y=0}^{y_2} \int_{x=1}^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy$$

$$= 5/24$$

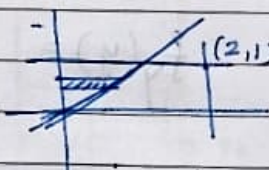
$$\textcircled{4} \quad P(x > 1 | y < y_2) = \frac{P(x > 1, y < y_2)}{P(y < y_2)} = \frac{5/24}{1/4}$$

$$= \frac{5}{24} \cdot \frac{4}{1} = \frac{20}{24}$$

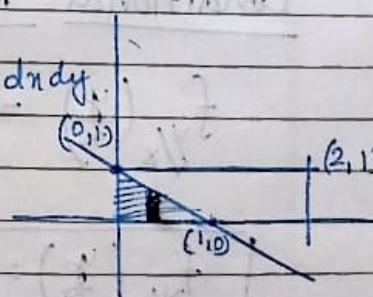
$$\textcircled{5} \quad P(y < y_2 | x > 1) = \frac{P(y < y_2, x > 1)}{P(x > 1)} = \frac{5/24}{19/24}$$

$$= \frac{5}{19}$$

$$\textcircled{6} \quad P(x < y) = \int_{y=0}^1 \int_{x=0}^y \left(xy^2 + \frac{x^2}{8} \right) dx dy$$

$$= \frac{53}{480}$$


$$\textcircled{7} \quad P(x+y \leq 1) = \int_{x=0}^1 \int_{y=0}^{1-x} \left(xy^2 + \frac{x^2}{8} \right) dx dy$$

$$= \frac{13}{480}$$


Q:- Given joint probability function

$$f(x, y) = 4xy, \quad 0 < x < 1, \quad 0 < y < 1$$

$$= 0 \quad \text{o.w.}$$

① Find the marginal density function

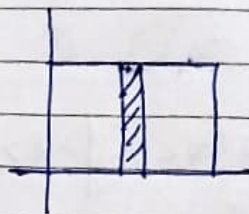
② conditional prob function

③ Are X and Y independent.

④ find $P(X < \frac{1}{2})$ and $P(X < \frac{1}{2}, Y < \frac{1}{2})$.

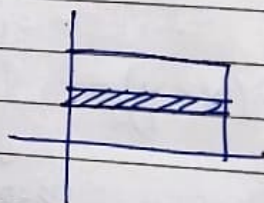
Soln:- $f_x(x) = \int_y f(x, y) dy$

$$f_x(x) = \int_0^1 4xy dy = 2x$$



$$\boxed{f_x(x) = 2x} \quad 0 < x < 1$$

$$f_y(y) = \int_0^1 4xy dx$$



$$\boxed{f_y(y) = 2y}, \quad 0 < y < 1$$

conditional Probability function:-

$$f_{y/x}\left(\frac{y}{x}\right) = \frac{f(x, y)}{f_x(x)} = \frac{4xy}{2x} = 2y$$

$$\boxed{f_{y/x}\left(\frac{y}{x}\right) = 2y},$$

slly

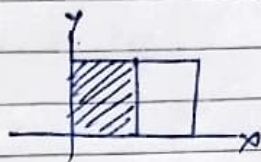
$$\boxed{f_{x/y}\left(\frac{x}{y}\right) = 2x}$$

Now $f_x(x) \cdot f_y(y) = 4xy$

$$= f(x, y)$$

ie x and y are independent.

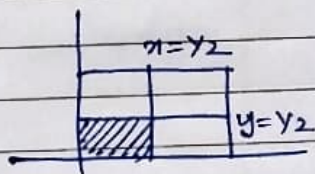
$$(4) \quad P(X < \frac{1}{2}) = \int_{x=0}^{\frac{1}{2}} \left(\int_{y=0}^{\frac{1}{2}} f(x, y) dy \right) dx$$



$$= \int_{x=0}^{\frac{1}{2}} f_x(x) dx = \int_0^{\frac{1}{2}} 2x dx = \frac{1}{4}$$

$$\left[P(X < \frac{1}{2}) = \frac{1}{4} \right]$$

$$(5) \quad P(X < \frac{1}{2}, Y < \frac{1}{2}) = \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} 4xy dx dy$$



Q The joint P.d.f of R.v (X, Y) is given by

$f(x, y) = Kxy e^{-(x^2+y^2)}$, $x > 0, y > 0$. Find the value of K and prove that X and Y are independent.

Soln:-

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\int_0^{\infty} \int_0^{\infty} Kxy e^{-(x^2+y^2)} dx dy = 1$$

$$\int_0^{\infty} y e^{-y^2} \left(\int_0^{\infty} x e^{-x^2} dx \right) dy = 1$$

$$= \int_0^{\infty} y e^{-y^2} \left(\frac{1}{2} \int_0^{\infty} e^{-t} dt \right) dy = 1$$

$$\begin{aligned} x^2 &= t \\ 2x dx &= dt \end{aligned}$$

$$= \frac{1}{2} \int_0^{\infty} y e^{-y^2} dy = \frac{1}{4} K = 1$$

$$\Rightarrow \boxed{K=4}$$

$$f_X(x) = \int_{y=0}^{\infty} 4xy e^{-x^2} e^{-y^2} dy$$

$$= 2x e^{-x^2} \quad x > 0$$

$$f_Y(y) = 2y e^{-y^2} \quad y > 0$$

$$f_x(x) \times f_y(y) = f_{xy}(x, y)$$

ie x and y are independent.

Ex:- $f_{xy}(x, y) = cx(x-y), 0 < x < 2, -x < y < x$

0 0.0

(a) find c (b) find $f_x(x)$

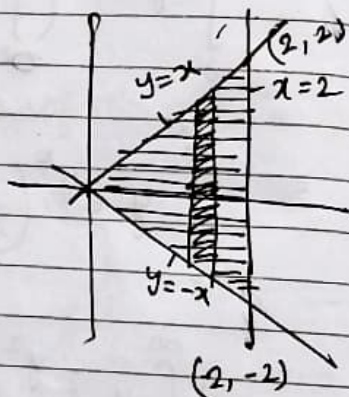
(c) $f_{y/x}(\frac{y}{x})$ (d) $f_y(y)$

Soln:-

(a) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

$$\int_0^2 \int_{-x}^x cx(x-y) dy = 1$$

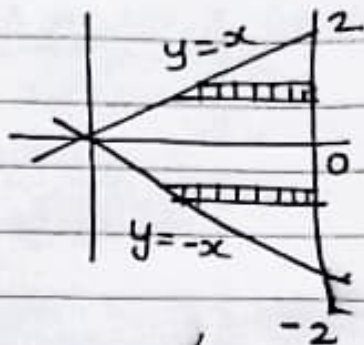
$$8c = 1 \Rightarrow \boxed{c = 1/8}$$



(b) $f_x(x) = \int_{y=-x}^x f(x, y) dy = \frac{x^3}{4}, 0 < x < 2$

(c) $f_{y/x}(\frac{y}{x}) = \frac{f(x, y)}{f_x(x)} = \frac{1}{2x^2}(x-y), 0 < x < 2, -x < y < x$

$$(d) f_y(y) = \begin{cases} \int_{x=-y}^2 f(x,y) dx, & -2 \leq y \leq 0 \\ \int_{x=y}^2 f(x,y) dx, & 0 \leq y \leq 2 \end{cases}$$



$$f_y(y) = \begin{cases} \int_{x=-y}^2 \frac{1}{8} x(x-y) dx, & -2 \leq y \leq 0 \\ \int_{x=y}^2 \frac{1}{8} x(x-y) dx, & 0 \leq y \leq 2 \end{cases}$$

$$f_y(y) = \begin{cases} \frac{1}{8} \left[\frac{8}{3} - 2y + \frac{5y^3}{6} \right], & -2 \leq y \leq 0 \\ \frac{1}{8} \left[\frac{8}{3} - 2y + \frac{y^3}{6} \right], & 0 \leq y \leq 2 \end{cases}$$

Expected value of two dimensional R.V.:-

The mathematical expectation of $g(x, y)$ of two dimensional random variable with joint p.m.f p_{ij} (discrete case) or joint p.d.f $f(x, y)$ (cont. case), is given by.

$$E[g(x, y)] = \sum_j \sum_i g(x_i, y_j) p_{ij}$$

$$E[g(x, y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) \cdot f(x, y) dx dy$$

Prop. of expectation:-

$$(1) E(X+Y) = E(X) + E(Y)$$

pf: $E(X+Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f(x, y) dx dy$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy$$

$$[E(X+Y) = E(X) + E(Y)]$$

Generalization

$$(1) E(X_1 + X_2 + X_3 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

(2) Multiple theo of Expectation:-

If x and y are independent random var. then

$$E(xy) = E(x) \cdot E(y)$$

Pf:- $E(xy) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (xy) f(x, y) dx dy$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (xy) f(x) \cdot f(y) dx dy$$

as x & y are independent

$$= \int_{-\infty}^{\infty} x f(x) dx \int_{-\infty}^{\infty} y f(y) dy$$

$$\boxed{E(xy) = E(x) \cdot E(y)}$$

Similarly $[E(x_1 \cdot x_2 \cdot x_3 \dots x_n) = E(x_1) E(x_2) \dots E(x_n)]$

③ $E(ax+b) = aE(x) + b$

$$E(ax+b) = \int_{-\infty}^{\infty} (ax+b) f(x) dx$$

$$= a \int_{-\infty}^{\infty} x f(x) dx + b \int_{-\infty}^{\infty} f(x) dx$$

$$= a E(x) + b \quad \text{as } \int_{-\infty}^{\infty} f(x) dx = 1$$

④ (a) $E\left(\frac{1}{x}\right) \neq \frac{1}{E(x)}$ (e) $E(a) = a$

$a = \text{constant}$

(b) $E(\sqrt{x}) \neq (E(x))^{1/2}$

(c) $E(\log x) \neq \log(E(x))$

(d) $E(x^2) \neq [E(x)]^2$

Variance:-

$$E[x - E(x)]^2 = E(x^2) - [E(x)]^2$$

One dim r.v.

Prop:-

(i) $V(ax+b) = a^2 V(x)$

(ii) $V(b) = 0$, variance of const. is zero.

(iii) $\text{Var}(x+y) = \text{Var}(x) + \text{Var}(y) + 2 \text{cov}(x, y)$

Covariance:-

If x and y are two r.v.s, then covariance between them is define as

$$\text{COV}(x, y) = E[(x - E(x))(y - E(y))]$$

$$= E(xy - xE(y) - yE(x) + E(x) \cdot E(y))$$

$$= E(xy) - E(x) \cdot E(y) - E(y) \cdot E(x) + E(x) \cdot E(y)$$

$$\boxed{\text{COV}(x, y) = E(xy) - E(x) \cdot E(y)}$$

$$[\text{COV}(x, y) = \frac{1}{n} \sum x_i y_i - \frac{1}{n} \sum x_i \cdot \frac{1}{n} \sum y_i] \quad \text{for discrete r.v.}$$

* If x and y are indep. r.v.s then

$$E(xy) = E(x) \cdot E(y)$$

then $\boxed{\text{COV}(x, y) = 0}$

Prop:- $\text{COV}(ax, by) = E[(ax - E(ax))(by - E(by))]$

$$= E[a(x - E(x)) \cdot b(y - E(y))]$$

$$= ab E[(x - E(x))(y - E(y))]$$

$$\boxed{\text{COV}(ax, by) = ab \text{COV}(x, y)}$$

$$(2) \text{COV}(X+a, Y+b) = \text{COV}(X, Y)$$

B) If X and Y are independent r.v then
 $E(XY) = E(X) \cdot E(Y)$

ie

$$\boxed{\text{COV}(X, Y) = 0}$$

$$(4) \text{COV}\left(\frac{X-\bar{X}}{\sigma_X}, \frac{Y-\bar{Y}}{\sigma_Y}\right) = \frac{1}{\sigma_X \sigma_Y} \text{COV}(X, Y)$$

$$(5) \text{COV}(aX+b, cY+d) = ac \text{COV}(X, Y)$$

$$(6) \text{COV}(X, Y+Z) = \text{COV}(X, Y) + \text{COV}(X, Z)$$

(7)

for two dimensional random variables X & Y
 Let $f(x, y)$ be joint p.d.f.

Let $f_X(x)$ be marginal prob. dis of X

& $f_Y(y)$ be marginal prob dis of Y

then,
$$\boxed{E(X) = \int x f_X(x) dx}$$

$$\boxed{E(Y) = \int y f_Y(y) dy}$$

Q:- Let $\phi(x, y) = 2x^2 + xy$, x .

Q: Two r.v.s x and y have joint P.d.f

$$f(x, y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4, 1 < y \leq 5 \\ 0, & \text{o.w.} \end{cases}$$

Find ① $E(x)$ ② $E(y)$ ③ $E(xy)$ ④ $E(2x+3y)$
⑤ $\text{Var}(x)$ ⑥ $\text{Cov}(x, y)$

Soln:- $E(x) = \int_{x=0}^4 x f_x(x) dx$

$$= \int_0^4 \frac{x^2}{8} dx$$

$$E(x) = \int_0^4 \frac{x^2}{8} dx = \frac{(x^3)}{8 \times 3} \Big|_0^4$$

$$= \frac{1 \times 4^3}{8 \times 3}$$

$$= \frac{64}{24 \times 3}$$

$$\boxed{E(x) = \frac{8}{3}}$$

$$f_x(x) = \int_1^5 f(x, y) dy$$

$$= \int_1^5 \frac{xy}{96} dy$$

$$f_x(x) = \left[\frac{y^2}{2} \right]_1^5 \frac{x}{96}$$

$$f_x(x) = \frac{x}{96} \times \frac{24}{2}$$

$$\boxed{f_x(x) = \frac{x}{8}}$$

$$f_y(y) = \int_0^4 f(x, y) dx$$

$$= \int_0^4 \frac{xy}{96} dx$$

$$= \frac{y}{96} \left[\frac{x^2}{2} \right]_0^4$$

$$= \frac{y}{96} \times \frac{16}{2}$$

$$\boxed{f_y(y) = \frac{y}{12}}$$

$$(2) E(Y) = \int_1^5 \int_0^4 y \left(\frac{xy}{96} \right) dx dy \text{ or } \int_1^5 y \cdot f_y(y) dy$$

$$E(Y) = \frac{31}{9}$$

$$(3) E(XY) = \int_1^5 \int_0^4 (xy) \frac{xy}{96} dx dy$$

$$E(XY) = \frac{248}{27}$$

$$(4) E(2X + 3Y) = 2E(X) + 3E(Y)$$

$$= 2 \times \frac{8}{3} + 3 \times \frac{31}{9} = \frac{47}{3}$$

$$(5) \text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_1^5 \int_0^4 x^2 \cdot \left(\frac{xy}{96} \right) dx dy$$

$$E(X^2) = 8$$

$$\text{Var}(X) = 8 - \left(\frac{8}{3} \right)^2 = \frac{8}{9}$$

$$(6) \text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

$$= \frac{248}{27} - \left(\frac{8}{3} \right) \left(\frac{31}{9} \right)$$

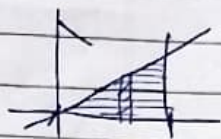
$$\boxed{\text{Cov}(X, Y) = 0}$$

Q. Let X and Y be continuous r.v. with joint p.d.f $f_{xy}(x, y) = 3x, 0 \leq y \leq x \leq 1$ and zero otherwise. (a) Find the covariance between X and Y . (b) Find Marginal p.d.f.s, expectations and variances of X and Y .

Solⁿ:

Marginal p.d.f of x

$$f_x(x) = \int_y f(x, y) dy$$



$$\left[f_x(x) = \int_{y=0}^x 3x dy = 3x^2, 0 \leq x \leq 1 \right]$$

Marginal p.d.f of y

$$f_y(y) = \int_{x=-\infty}^{\infty} f(x, y) dx$$



$$= \int_{x=y}^1 3x dx = \frac{3}{2}(1-y^2), \quad 0 \leq y \leq 1$$

$$\left[f_y(y) = \frac{3}{2}(1-y^2), 0 \leq y \leq 1 \right]$$

Expectations:- $E(X) = \int_{x=-\infty}^{\infty} x f_x(x) dx$

$$= \int_{x=0}^1 x \times 3x^2 dx = \frac{3}{4}$$

$$\left[E(X) = \frac{3}{4} \right]$$

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$$E(X^2) = \int_0^1 x^2 \cdot f_X(x) dx = \frac{3}{5}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{3}{80} \end{aligned}$$

Now

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y \cdot f_Y(y) dy \\ &= \int_0^1 \frac{3}{2} y (1-y^2) dy \end{aligned}$$

$$[E(Y) = \frac{3}{8}]$$

$$E(Y^2) = \int_0^1 y^2 \cdot \frac{3}{2} (1-y^2) dy = \frac{1}{5}$$

$$\begin{aligned} \text{Var}(Y) &= E(Y^2) - [E(Y)]^2 \\ &= \frac{1}{5} - \left(\frac{3}{8}\right)^2 = \frac{19}{320} \end{aligned}$$

Now

$$\text{cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x,y) dy dx$$

$$= \int_{-\infty}^{\infty} = \int_0^1 \int_0^x xy \cdot 3x \, dy \, dx$$

$$E(xy) = \frac{3}{2} \left(\frac{x^5}{5} \right)_0^1 = \frac{3}{10}$$

Now $\text{cov}(x, y) = E(xy) - E(x) \cdot E(y)$

$$= \frac{3}{10} - \frac{3}{4} \times \frac{3}{8} = \frac{3}{160}$$

$$\boxed{\text{cov}(x, y) = \frac{3}{160}}$$

Q If the joint density for the random variable (X, Y) where X is the unit temperature change and Y is the proportion of spectrum shift that a certain atomic particle produce is given by

$$f(x, y) = \begin{cases} cxy^2, & 0 < x < y < 1 \\ 0, & \text{o.w} \end{cases}$$

find

- (i) value of c (ii) $E(XY)$ (iii) $f_X(x)$ (iv) $f_Y(y)$
(v) $f_{Y/X}(y/x)$

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correlation coefficient:-

If x and y are two r.v.s then
coeff. of correlation b/w x and y
is defined as r_{xy} or ρ_{xy} , which
is a numerical measure of linear
relationship b/w them

$$r_{xy} = \frac{\text{cov}(xy)}{\sigma_x \sigma_y} = \frac{\text{cov}(xy)}{\sqrt{\text{var}(x)} \sqrt{\text{var}(y)}}$$

Prop. ① If x and y are independent

$$\text{cov}(xy) = 0$$

$$\text{ie } \boxed{r_{xy} = 0}$$

but the converse is not true.
when $r_{xy} = 0$, we say that x and
 y are uncorrelated.

② $|r_{xy}| \leq 1$

if $r_{xy} = +1$ ie +ive correlation
 $r_{xy} = -1$ ie -ive correlation

$r_{xy} = 0$, uncorrelated r.v.s